

EXO UML practice

Task 1

Class Patient

→ Attributes:
patient_id: int
name: str
email: str
password: str

→ Methods:
submit_problem()
receive_prescription
make_payment

Class Doctor

→ Attributes:
doctor_id: int
name: str
specialty: str
password: str

→ Methods:
review_problem()
write_prescription
receive_payment

Class organizer

→ Attributes:
organizer_id
name: str
role: str
password: str

→ Methods:
assign_doctor()
send_prescription()
forward_payment

Relationships

Patient 0..* --- 1 Organizer

Organizer 1 --- 0..* Doctor

Patient 0..* --- 0..* Doctor

Task 2

CLASS HealthProblem

Attributes

- problem_id: int
- description: str
- created_at: datetime
- status

Methods

- update_status()
- assign_doctor()
- mark_resolved()

CLASS Prescription

Attributes

- prescription_id
- content
- issued_at
- is_reusable

Methods

- generate()
- resend()
- archive()

Class Payment

Attributes

- payment_id: int
- amount: float
- date
- status: str

Methods

- process()
- validate()
- refund()

Relationships:

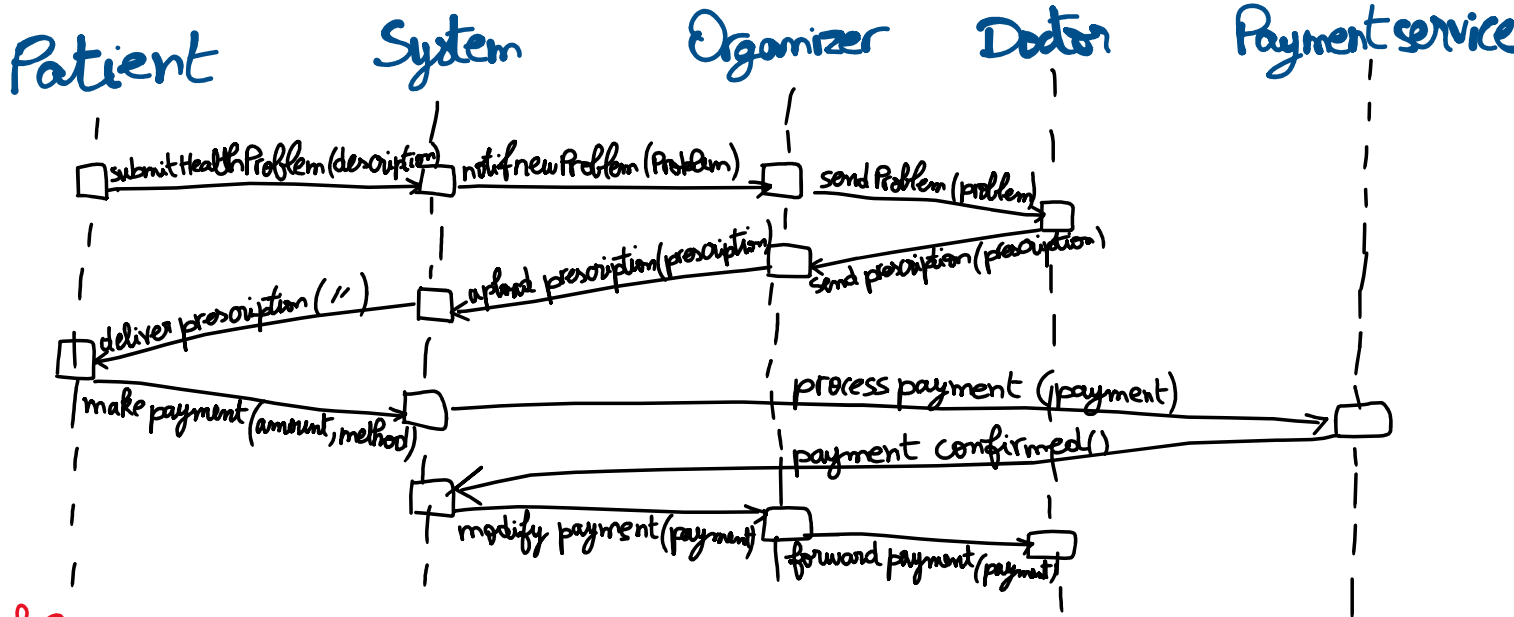
Patient "1" --- "0..*" HealthProblems
Organizer "1" --- "0..*" HealthProblem
Doctor "0" .. --- "0..*" HealthProblem
HealthProblem "1" --- "0..1" Prescription
Patient "1" --- "0..*" Payment
Organizer "1" --- "0..*" Payment
Doctor "1" --- "0..*" Payment

Payment inheritance

Payment

- Cash payment
- Credit payment
- Check payment
 ↑
 card

Sequence diagram



Task 3

Class User

ATTRIBUTES

- id : int
- name : String
- password : String

METHODS

- + login()
- + logout()
- + change password

Inheritance

USER

- Patient
- Doctor
- organizer

Past Prescription

Patient « 1 » --- « 0.. » Prescription

Status management

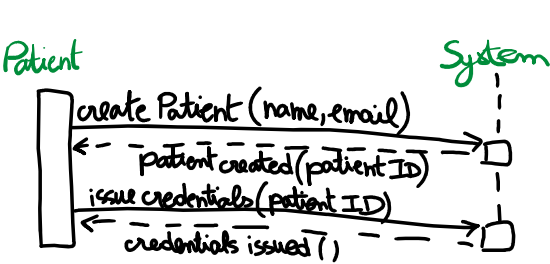
ENUM ProblemStatus

- Pending
- In_review
- Resolved

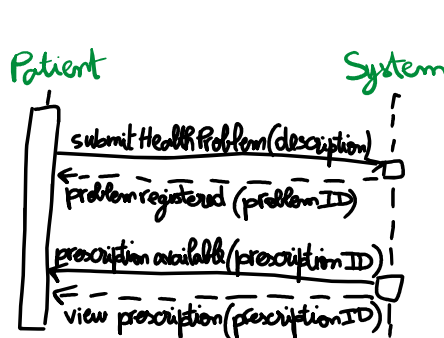
HealthProblem

- Status : problemStatus
- updatedAt
- resolvedAt

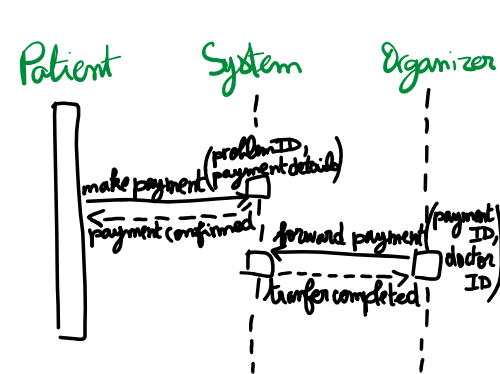
SYSTEM SEQUENCE DIAGRAM



Patient registration



Consultation



Payment

EXO 2

Task 1.1 : Use Case Diagram

Actors

- Member
- Order Processing Clerk

Use cases

- Place Order
- Verify Membership
- Process Order
- Make Payment

Relations

- Member → Place Order
- Member → Make Payment
- Order Processing Clerk → Verify Membership
- Order Processing Clerk → Process Order

Task 1.2 : Class Diagram

1. Member

Attributes

- memberId : int
- name : String
- address : String
- memberType : String

Methods

- placeOrder()
- makePayment()
- viewDiscount()

2. Order

Attributes

- orderId : int
- orderDate : Date
- status : String
- totalAmount : float

Methods

- calculateTotal()
- confirmOrder()
- saveToDailyRecord()

3. OrderProcessingClerk

Attributes

- clerkId : int
- name : String
- department : String
- employeeNumber : String

Methods

- verifyMembership()
- processOrder()
- checkAvailability()

Associations

- one **Member** places **many Orders**
- one **OrderProcessingClerk** processes **many Orders**

TASK 2.1

System: ABCD Records Ordering System

Actors:

- Member
- Non-Member
- Order Processing Clerk
- Collection Department Clerk

Use Cases:

- Place Order
- Verify Membership
- Verify Item Availability
- Process Order
- Apply Discount
- Print Invoice
- Print Shipping List
- Order Unavailable Items
- Make Payment
- Send Membership Application

Relationships:

- Place Order <<include>> Verify Membership
- Place Order <<include>> Verify Item Availability
- Place Order <<include>> Process Order
- Place Order <<include>> Print Invoice
- Place Order <<include>> Print Shipping List
- Apply Discount <<extend>> Place Order
- Order Unavailable Items <<extend>> Place Order
- Send Membership Application <<extend>> Verify Membership

TASK 2.2

CLASS Member

Attributes: memberId, name, address, email

Methods: placeOrder(), makePayment()

CLASS RegularMember (Member)

Attributes: discountRate

Methods: canOrderUnavailable()

CLASS RoyalMember (Member)

Attributes: discountRate, priorityLevel

Methods: canOrderUnavailable()

CLASS Order

Attributes: orderId, orderDate, status, totalAmount

Methods: calculateTotal(), saveRecord()

CLASS Item

Attributes: itemId, title, price, available

Methods: isAvailable()

CLASS CD (Item)

Attributes: artist

Methods: getType()

CLASS Tape (Item)

Attributes: duration

Methods: getType()

CLASS Invoice

Attributes: invoiceId, issueDate, total

Methods: printInvoice()

CLASS ShippingList

Attributes: shippingId, shipDate, address

Methods: printShippingList()

CLASS Payment

Attributes: paymentId, amount, method

Methods: process()

CLASS CashPayment (Payment)

Attributes: receiptNumber

Methods: process()

CLASS CheckPayment (Payment)

Attributes: checkNumber, bankName

Methods: process()

CLASS OrderLine

Attributes: quantity, unitPrice

Methods: lineTotal()

CLASS BankDraftPayment (Payment)

Attributes: draftNumber, bankName

Methods: process()

CLASS MembershipApplication

Attributes: applicationId, name, address, date

Methods: submit(), approve()

CLASS OrderProcessingClerk

Attributes: clerkId, name, employeeNumber

Methods: verifyMembership(), processOrder()

CLASS CollectionDepartmentClerk

Attributes: clerkId, name, employeeNumber

Methods: receivePayment(), validatePayment()

RELATIONSHIPS

Member 1 --- 0..* Order

Order 1 --- 1..* OrderLine

OrderLine 1 --- 1 Item

Order 1 --- 1 Invoice

Order 1 --- 1 ShippingList

Order 1 --- 1 Payment

